

FLOODGAUGE · DESIGN DECISIONS REGISTER

FloodGauge design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from REVIEW.md, FLG-DDR-001 to FLG-DDR-003 and the build plan work

FloodGauge design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Open decisions, all Proposed, awaiting Amish.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction changes (P1 to P11)	Accept; accept with changes; return to the concept	Accept	Every component and step of the build plan	FLG-DDR-003
2	Node height 3.4 m (was 3.0 m), so the street cable fits the 3 m I2C limit along its real route	Keep 3.4 m; keep 3.0 m with an I2C bus extender and a 4 m cable	Keep 3.4 m	FieldNode position on the pole (step 7); cable lengths	FLG-DDR-003, P6
3	R15 mass margin is 0.01 kg on estimated masses (4.99 kg against 5.0 kg)	Accept and weigh the prototype at TRL 4; lighten further now (for example a 2.5 mm bracket plate)	Accept and weigh at TRL 4	Bracket plate thickness	FLG-DDR-003, A1
4	Lifting the grate with the drain cable threaded through a grate opening	Undo the gutter cover's two anchor nuts and lift the grate with a 0.5 m slack loop in the basin; add an inline M12 joint at the riser guard foot	First option for the pilot; review after the first cleaning visit	Drain cable slack; maintenance	FLG-DDR-003, A2
5	R12 installation takes 120 min against 90 min on the surface route	Relax R12 to 120 min for the pilot route; keep 90 min and prefabricate the cover and bracket set	Relax R12 to 120 min for the pilot route	None (installation time only)	REVIEW.md, 2026-09-25, item 3; FLG-DDR-002, O3
6	Pilot partner and city for co-design and a first site; depth of the first alert band below 150 mm	Not yet identified	None made	Site pole, curb lane traffic, lens height within 2.5 to 5.0 m	FLG-DDR-001, O1; FLG-DDR-002, N5
7	Render layout of the pole kit drawn 3.3 m lower than installed	Keep as a render-only layout with a caption; render true heights	Keep, with the caption	Photoreal renders only	REVIEW.md, 2026-09-26, item 1
8	Drain kit shown only in the exploded render	Accept; add a cutaway render through the basin	Accept	Photoreal renders only	REVIEW.md, 2026-09-26, item 2
9	Side gland on the radar head and the street cable route	Adopt (now in the model, FLG-DDR-003 P5)	Adopt	Radar head housing (section 3.8 of the build plan)	REVIEW.md, 2026-09-26, item 3
10	Appearance details not in the BOM (clear lid window, name plate, status light, cover ridges)	Render detail only; raise a clear-lid node in the FieldNode repo	Render detail only	None	REVIEW.md, 2026-09-26, item 4

#	Decision needed	Options	Recommendation	Affects in the build	Source
11	Alert governance: who receives alerts, who acts, how residents opt in and out	To be set with the partner city	None made	Alert service only	FLG-PRC-001, open questions
12	Theft and vandalism protection with the node at 3.4 m and the marker and guard at street level	Tamper-resistant screw heads; leave as designed for the pilot	None made	Fixings on the marker bands and covers	FLG-PRC-001, open questions

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Radar module data sheet: accuracy, current, beam width with the lens, price and 60 GHz certification for fixed outdoor use in the pilot country	The depth error, energy and false-reading figures assume them	FLG-CAL-001, Table 1
2	The radar housing's lid boss pattern and a base that can take a 46 mm window	Sets the four M4 holes in the head plate and the lens fixing	FLG-DDR-003, P4
3	The ranger's transducer face seal (IP67 only) and potting compatibility	R9 is at risk on the face seal	FLG-REQ-001, R9
4	Band clamps that close on a 60 mm pole with the V-block and plate inside (about 330 to 380 mm round for the bracket and FieldNode, about 450 mm for the marker), and the torque that gives 1,000 N of preload	The twist and pull figures assume 1,000 N per band	FLG-CAL-001, H2 and H4
5	The site pole's wall thickness (2.5 mm assumed) and the owner's permission to drill it	Bearing strength of the through-bolt	FLG-CAL-001, H2b
6	A grate opening within 70 mm of the grate's curb edge on the chosen cover line, and the curb height (150 mm assumed)	The cable comes up through that opening; the curb cover length follows the curb	FLG-DDR-003, P7 and P8
7	The 75 mm socket coupling fits the chosen pipe, and the stand-off pipe clamps reach 90 mm from the wall	Sets the drain head fit and the tube position	FLG-DDR-003, P9 and P10
8	Outdoor cable mass and diameter (0.07 kg/m, 10 mm assumed)	Mass margin and the fit inside the 16 mm covers	FLG-CAL-001, I1

■ Value engineering

Value-engineering target: USD 160 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 190.50 for the FloodGauge-specific parts (USD 30.50 over the target); USD 329.50 with the USD 139.00 FieldNode core, which is costed in its own repo.

Main cost drivers: the street radar head (USD 55.00), the sensor arm and pole bracket (USD 35.00), the surface covers and riser guard (USD 26.00) and the drain head (USD 24.50).

Savings worth trying:

- Quote the radar module and a housing with the lens window already moulded; the USD 55.00 is unverified and is the largest line.
- Cut the bracket plate, cleats, clips and head plate from one sheet and one angle offcut, and buy band clamps and M6 fixings in packs shared with FieldNode builds.
- Make the riser guard and covers from one 2 mm galvanized strip length, or use a bought steel cable protector for the sidewalk run.
- Use the conduit route (about USD 8 plus civil work, not in this total) at sites where road works are planned anyway.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D9: FieldNode costed in its own repo; street and drain heads on one node; radar street head with ultrasonic as the lower-cost variant; ultrasonic drain head in a slotted stilling tube with a wet probe; arm over the gutter; 150 and 300 mm alert bands plus drain-full; private or city gateway for events; levels only; pitch and problem unchanged	Amish: "i accept all your recommendations, go with them across all repos."	FLG-DDR-001, FLG-DDR-002
2026-09-25	TRL 3 items N1 to N5: R3 restated; lens set per site within 2.5 to 5.0 m (4.6 m reference); surface cable route for pilots with conduit at permanent sites; M8 anti-rotation through-bolt; a first alert band below 150 mm supported	Amish, same instruction	FLG-DDR-002
2026-09-26	<code>budget_usd</code> set to USD 160	Amish: "i approve all the budget items."	FLG-DDR-002
2026-10-01	Budgets are value-engineering targets, not limits; cost is reported over or under the target	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	STANDARDS section 18
2026-10-01	Design for construction, P1 to P11	Made under Amish's 2026-09-30 instruction to make the design physically buildable ("fix the design assumptions to match and be physically feasible"); open for his review (open decision 1)	FLG-DDR-003